

Python Practice Handout

Variables, Integers and Floats

1. What is a variable

A variable is a name that holds a piece of information. That information can change over time. Variables can hold numbers or text.

Type this to store how much money is in your wallet:

```
wallet = 41
```

2. Creating a variable shows nothing

Run the program above. Nothing appears on the console. Creating a variable stores the value, but it does not display it.

```
wallet = 41
```

Console output: (nothing)

3. Printing a variable

To see what a variable is holding, you must print it. Put the variable name inside the parentheses, with no quotation marks.

```
wallet = 41  
  
print(wallet)
```

Output:

```
41
```

4. Blank lines do not matter

Python lets you put as many or as few blank lines between your statements as you like. All three of these run exactly the same way. Use whatever spacing is comfortable to read.

No blank line

```
wallet = 41  
print(wallet)
```

One blank line

```
wallet = 41  
  
print(wallet)
```

Several blank lines

```
wallet = 41

print(wallet)
```

5. Changing the value

The value inside a variable can change. Suppose you buy pizza and now have only \$32 left. Assign the new value to the same variable and print it again.

```
wallet = 41

print(wallet)

wallet = 32

print(wallet)
```

Output:

```
41
32
```

6. Try it yourself

Make a variable called day and set it equal to today's date of the month. Then print it. Try this on your own before looking below.

```
day = 21

print(day)
```

7. Types: int and float

A type is the kind of thing you put inside a variable. Two of the number types in Python are int and float. An int is a whole number, and a float is a number with a decimal.

8. Integers (int)

Int is short for integer. It means a whole number, with no decimal places attached to it. The variable day from earlier is an integer.

```
day = 21
```

Integers can be positive or negative. On a very cold day the temperature might be minus 15 degrees.

```
temp = -15
```

9. Floats

A float is a number that has a decimal place attached to it. If you track your weight precisely, that .4 matters, so you store it as a float.

```
weight = 190.4
```

You can have more than one number after the decimal, and floats can be positive or negative too.

```
bitcoin = 0.00435  
balance = -12.75
```

10. Doing math

Python can do calculations for you. Type the sum straight inside the print parentheses.

```
print(3 + 6)
```

Output:

```
9
```

11. Math with variables

You can also do math using the variables you have made. To find the date three days from now, add 3 to day.

```
day = 21  
  
print(day + 3)
```

Output:

```
24
```

These are the four basic operators. Multiplication uses a star, and division uses a forward slash.

```
weight = 190.4  
  
print(weight + 10)  
print(weight - 10)  
print(weight * 2)  
print(weight / 2)
```

Output:

```
200.4  
180.4  
380.8  
95.2
```

12. Try it yourself

Find something around you that you can count with a whole number, and something you would measure with a decimal. Put each one in a variable. Try this on your own before looking below.

```
age = 32  
  
bitcoin = 0.00435
```